operates according to a program that determines definitively which color to flash depending on which door is opened. If she and Mulder independently and randomly choose which of the three doors on each of their boxes to open, and then compare the colors they see—box by numbered box—they must find agreement for *more* than 50 percent of the boxes.

When cast in the language of physics, as it will be in the next section, Mulder's realization is nothing but John Bell's breakthrough.

Counting Angels with Angles

The translation of this result into physics is straightforward. Imagine we have two detectors, one on the left side of the laboratory and another on the right side, that measure the spin of an incoming particle like an electron, as in the experiment discussed in the section before last. The detectors require you to choose the axis (vertical, horizontal, back-forth, or one of the innumerable axes that lie in between) along which the spin is to be measured; for simplicity's sake, imagine that we have bargain-basement detectors that offer only three choices for the axes. In any given run of the experiment, you will find that the incoming electron is either spinning clockwise or counterclockwise about the axis you selected.

According to Einstein, Podolsky, and Rosen, each incoming electron provides the detector it enters with what amounts to a program: Even though it's hidden, even though you can't measure it, EPR claimed that each electron has a definite amount of spin—either clockwise or counterclockwise—about each and every axis. Hence, when an electron enters a detector, the electron definitively determines whether you will measure its spin to be clockwise or counterclockwise about whichever axis you happen to choose. For example, an electron that is spinning clockwise about each of the three axes provides the program *clockwise, clockwise, clockwise;* an electron that is spinning clockwise about the first two axes and counterclockwise about the third provides the program *clockwise, clockwise, counterclockwise,* and so forth. In order to explain the correlation between the left-moving and right-moving electrons, Einstein, Podolsky, and Rosen simply claim that such electrons have